

Script Housekeeping monitor:-

→ Repo link: []

→ Instructions to run the code is described in README.md

→ Function based design is preferred to retain simplicity.

→ logic: Read json → parse inputs → Apply threshold → Generate events → Write log

→ In the operational system, the threshold for telemetry parameter will be supplied by spacecraft thermal, avionics, storage, navigation and FDIR requirements. Currently, these values are assumed to run the code.

Improvements:-

- 1] If data is received at regular frequency, then one can set up the "cron job" to run the script & link it to "Grafana dashboard".
- 2] Python class based approach can be implemented for code readability.
- 3] With docker containerization, one can mimic the limited resource performance of the code.
- 4] Define the threshold rules for all parameter.

Question:-

1] Why not receive the house keeping telemetry in the format similar to TLE?

① It reduces the data transmission and also easy to handle with help of reference placeholders.

→ flexibility to include additional rules is provided in "threshold-check.py"

→ printing log can be improved using better logging tools. Yet to explore.

→ With meaningful threshold value, server the code to get proper insights.

→ Currently, code run on loops. With modification to input structure [similar to TLE], one can explore the option of dataframe and vectorize the operations.

→ Since threshold varies wot modes, its required to create multiple threshold files for every mode & compare from it.

→ In general, the input/output folder won't be pushed to GitHub. currently, they are pushed to simplify the evaluation.

— * — ↓
* Modify the code based on the feedback received.

→ Bug fixed: "write_event_log" file bypassing CLI input is found.

(.) filename name was hardcoded.

(.) It is removed & Readme.md modified.

C) pushed to github.

WIR → Write error handling, testcases, CCSDS packet protocol & FDIR scheme.

① Order to address.

1] FDIR
2] task scheduler
3] error handling
4] test cases.
↓
try ←
CCSDS
Python
package. [low priority]

↓
Modify the hk.jsm
to CCSDS compatible
format:-

② primary header
secondary header
and parse it within
the code.